

The nearly shortest path embedding in Crossed Cubes *

Pao-Lien Lai^{‡†}, Hong-Chun Hsu[‡]

[‡]Department of Computer Science and Information Engineering
National Dong Hwa University, Shoufeng, Hualien, Taiwan 97401, R.O.C.

[‡]Department of Medical Informatics,
Tzu Chi University, Hualien, Taiwan 970, R.O.C.

Abstract

A graph G is *panconnected* if each pair of distinct vertices $u, v \in V(G)$ are joined by a path of length l with $d_G(u, v) \leq l \leq |V(G)| - 1$, where $d_G(u, v)$ is the length of shortest path joining u and v in G . Fan et. al.[7] proved that n -dimension Crossed cube, CQ_n , is almost panconnected except the path of length $d_{CQ_n}(u, v) + 1$ for any two distinct vertices $u, v \in CQ_n$ recently. In this paper, We give an construction scheme to construct the path of length $d_{CQ_n}(u, v) + 1$ for any two distinct vertices $u, v \in CQ_n$ in some conditions. Moreover, we give the necessary and sufficient conditions to check the existence of the path of length $d_{CQ_n}(u, v) + 1$ for any two distinct vertices u, v in CQ_n .

Keywords: Interconnection network; Crossed cube; Panconnectivity; Path embedding.

1 Introduction

For the graph definition and notation we follow [2]. $G = (V, E)$ is a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the *vertex set* and E is the *edge set*. Two vertices u and v are *adjacent* if $(u, v) \in E$. A path is a sequence of adjacent vertices, written as $\langle v_0, v_1, v_2, \dots, v_m \rangle$, in which all the vertices $v_0, v_1, \dots, v_m$ are distinct except possibly $v_0 = v_m$. We also write the path $\langle v_0, P, v_m \rangle$, where $P = \langle v_1, \dots, v_{m-1} \rangle$. The *length* of a path P , denoted by $\text{len}(P)$, is the number of edges in P . The *distance* between u and v in G , denoted

by $d_G(u, v)$, is the length of the shortest path of G joining u and v .

An n -dimensional crossed cube, CQ_n , is a variation of hypercube Q_n and preserves many of its desirable properties. CQ_n has 2^n vertices and $n2^{n-1}$ links, same as hypercube Q_n . However, it has a smaller diameter $\lceil \frac{n+1}{2} \rceil$, about half of that of Q_n . Linear arrays and rings are two of the most primary networks for parallel and distributed computation. They are both favorable for developing simple algorithms with low communication costs. Many efficient algorithms designed on linear arrays and rings for solving a variety of algebraic problems and graph problems can be found in previous works [1, 10].

The path embedding problem, which deals with all the possible lengths of the paths, is studied in many papers of Broersma [3], Kanetkar [8], and Seng et al.[12]. For this definition, *panconnected* is proposed. A graph G is *panconnected* if each pair of distinct vertices u and v are joined by a path of length l , for all $d_G(u, v) \leq l \leq |V(G)| - 1$. A graph G is *nearly panconnected* if each pair of distinct vertices u and v are joined by a path of length l , for $l = d_G(u, v)$ and for all $d_G(u, v) + 2 \leq l \leq |V(G)| - 1$. CQ_n is nearly panconnected that is showed in [7]. In this paper, We give a scheme to construct the path of length $d_{CQ_n}(u, v) + 1$ for any two distinct vertices $u, v \in CQ_n$ in some conditions. Moreover, we give the necessary and sufficient conditions to check the existence of the path of length $d_{CQ_n}(u, v) + 1$ for any two distinct vertices u, v in CQ_n .

The rest of this paper is organized as follows. In next section, we discuss the definition and some properties of Crossed cubes. In section 3, we give the construction scheme for nearly shortest path if it exists or else we prove it doesn't exist. In the final section, we give the conclusion.

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[†]Correspondence to: Assistant Professor Pao-Lien Lai, Department of Computer Science and Information Engineering, National Dong Hwa University, Shoufeng, Hualien, Taiwan 97401, R.O.C. e-mail: baolein@mail.ndhu.edu.tw

2 Crossed Cubes

To define Crossed Cubes, we follow the Yang's definition [13]. First we need the following pair-related mapping definition.

Definition 1 Let n be a positive integer. Let $\{0, 1\}^n$ denote the set of all binary strings with 0, 1 of length n . A pair-related mapping of order n , denoted by PR_n , is a bijective mapping from $\{0, 1\}^n$ to $\{0, 1\}^n$ which is recursively defined as follows.

- (1) For $n = 1$, $PR_1(0) = 0$, $PR_1(1) = 1$.
- (2) For $n = 2$, $PR_2(00) = 00$, $PR_2(10) = 10$, $PR_2(01) = 11$, $PR_2(11) = 01$.
- (3) For odd $n \geq 3$, $PR_n(u_{n-1}u_{n-2} \dots u_1u_0) = PR_1(u_{n-1})PR_{n-1}(u_{n-2}u_{n-3} \dots u_1u_0)$.
- (4) For even $n \geq 4$, $PR_n(u_{n-1}u_{n-2} \dots u_1u_0) = PR_2(u_{n-1}u_{n-2})PR_{n-2}(u_{n-3}u_{n-4} \dots u_1u_0)$.

The following is the definition of the n -dimensional Crossed cube CQ_n .

Definition 2 Let $n \geq 2$ be an integer. An n -dimensional Crossed Cube, denoted by CQ_n , is a graph with vertex set $V(CQ_n) = \{0, 1\}^n$. Two vertices $u = u_{n-1}u_{n-2} \dots u_1u_0$ and $v = v_{n-1}v_{n-2} \dots v_1v_0$ are adjacent if and only if there is an index k such that

- (1) $u_i = v_i$ for each $k+1 \leq i \leq n-1$,
- (2) $u_k = 1 - v_k$,
- (3) $PR_k(u_{k-1}u_{k-2} \dots u_1u_0) = v_{k-1}v_{k-2} \dots v_1v_0$.

For convenience, we use $(u)^i$ to represent the i -th neighbor of vertex u where the left-first different bit between u and $(u)^i$ is i -th bit. That is, $u_j = ((u)^i)_j$ for $j > i$ and $u_i = 1 - ((u)^i)_i$. For the definition of the pair-related mapping and Crossed Cubes, we have the following properties.

Property 1 $PR_k(PR_k(u_{k-1}u_{k-2} \dots u_1u_0)) = u_{k-1}u_{k-2} \dots u_1u_0$

Property 2 Let $Z = \{00, 10\}^k$, then $PR_{2k}(Z) = Z$.

3 The existence of the nearly shortest path

Given any pair of double-bits, we get the set $U = \{(00, 00), (00, 01), (00, 10), (00, 11), (01, 00), (01, 01), (01, 10), (01, 11), (10, 00), (10, 01), (10, 10), (10, 11), (11, 00), (11, 01), (11, 10), (11, 11)\}$.

Definition 3 Let $EP = \{(00, 00), (10, 10)\}$. Two two-digit binary strings $u = u_1u_0$ and $v = v_1v_0$ are an equal pair if and only if $(u, v) \in EP$.

Definition 4 Let $NEP = U/EP = \{(00, 01), (00, 10), (00, 11), (10, 00), (10, 01), (10, 11), (01, 00), (01, 10), (01, 11), (01, 01), (11, 00), (11, 10), (11, 01), (11, 11)\}$. Two two-digit binary strings $u = u_1u_0$ and $v = v_1v_0$ are a non-equal pair if and only if $(u, v) \in NEP$.

Let $u = u_{n-1}u_{n-2} \dots u_2u_1u_0$ and $v = v_{n-1}v_{n-2} \dots v_2v_1v_0$ be two distinct vertices of CQ_n . Let $(u_{2k+1}u_{2k}, v_{2k+1}v_{2k})$ be the right most double-bits pair belongs to NEP where $\lfloor (n-1)/2 \rfloor \geq k \geq 0$. Then we can partition u and v into three parts, respectively, $u = X\alpha_2\alpha_1Z$ and $v = Y\beta_2\beta_1Z$, where $\alpha_2\alpha_1 = u_{2k+1}u_{2k}$, $\beta_2\beta_1 = v_{2k+1}v_{2k}$, $|Z| = 2k$, and $|X| = |Y| = n - 2(k+1)$ for some $\lfloor (n-1)/2 \rfloor \geq k \geq 0$. For example, let $u = 01100101010$ and $v = 11001111010$ be two vertices of CQ_{11} then $Z = 1010$, $\alpha_2\alpha_1 = 10$, $\beta_2\beta_1 = 11$, $X = 01100$, and $Y = 11001$. Based on the shortest path routing algorithm of K. Efe [5, 6], suppose that $d_{CQ_{n-2(k+1)}}(X, Y) = m \geq 1$, we have the shortest path routing scheme listed in Table 1.

Lemma 1 Suppose that $u = X\alpha_2\alpha_1Z$ and $v = Y\beta_2\beta_1Z$ are two distinct vertices of CQ_n and $(\alpha_2\alpha_1, \beta_2\beta_1)$ be the right first double-bits pair belongs to NEP . If $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(01, 11), (11, 01)\}$, then $d_{CQ_n}(u, v)$ is odd.

Lemma 2 Suppose that $u = X\alpha_2\alpha_1Z$ and $v = Y\beta_2\beta_1Z$ are two distinct vertices of CQ_n and $(\alpha_2\alpha_1, \beta_2\beta_1)$ be the right first double-bits pair belongs to NEP . If $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(01, 01), (11, 11)\}$, then $d_{CQ_n}(u, v)$ is even.

Lemma 3 Suppose that $u = X\alpha_2\alpha_1Z$ and $v = Y\beta_2\beta_1Z$ are two distinct vertices of CQ_n such that $(\alpha_2\alpha_1, \beta_2\beta_1)$ is the right most double-bits pair belongs to NEP and $|Z| = 2k$ for some $k \geq 0$. Let P be a shortest path joining u and v in CQ_n . Then each edge, $(w, (w)^i)$, of P is with $2k \leq i \leq n-1$.

Table 1: The shortest path routing scheme.

$(\alpha_2\alpha_1, \beta_2\beta_1)$	Shortest Path Routing Scheme.
(00, 01)	$X00Z \xrightarrow{m-i \text{ steps}} X'00Z \rightarrow X'01Z \xrightarrow{i(\text{even}) \text{ steps}} Y01Z$ (With $i \geq 0$.)
(00, 10)	$X00Z \xrightarrow{m \text{ steps}} Y00Z \rightarrow Y10Z$
(00, 11)	$X00Z \xrightarrow{m-i \text{ steps}} X'00Z \rightarrow X'01Z \xrightarrow{i(\text{odd}) \text{ steps}} Y11Z$ (With $i \geq 1$.)
(01, 00)	$X01Z \xrightarrow{i(\text{even}) \text{ steps}} X'01Z \rightarrow X'00Z \xrightarrow{m-i \text{ steps}} Y00Z$ (With $i \geq 0$.)
(01, 01)	$X01Z \xrightarrow{m \text{ steps}} Y11Z \rightarrow Y01Z$ (When m is odd.) $X01Z \xrightarrow{m \text{ steps}} Y01Z$ (When m is even.)
(01, 10)	$X01Z \xrightarrow{i(\text{odd}) \text{ steps}} Y'11Z \rightarrow Y'10Z \xrightarrow{m-i \text{ steps}} Y10Z$ (With $i \geq 1$.)
(01, 11)	$X01Z \xrightarrow{m \text{ steps}} Y11Z$ (When m is odd.) $X01Z \xrightarrow{m \text{ steps}} Y01Z \rightarrow Y11Z$ (When m is even.)
(10, 00)	$X10Z \xrightarrow{m \text{ steps}} Y10Z \rightarrow Y00Z$
(10, 01)	$X10Z \xrightarrow{m-i \text{ steps}} X'10Z \rightarrow X'11Z \xrightarrow{i(\text{odd}) \text{ steps}} Y01Z$ (With $i \geq 1$.)
(10, 11)	$X10Z \xrightarrow{m-i \text{ steps}} X'10Z \rightarrow X'11Z \xrightarrow{i(\text{even}) \text{ steps}} Y11Z$ (With $i \geq 0$.)
(11, 00)	$X11Z \xrightarrow{i(\text{odd}) \text{ steps}} X'01Z \rightarrow X'00Z \xrightarrow{m-i \text{ steps}} Y00Z$ (With $i \geq 1$.)
(11, 01)	$X11Z \xrightarrow{m \text{ steps}} Y01Z$ (When m is odd.) $X11Z \xrightarrow{m \text{ steps}} Y11Z \rightarrow Y01Z$ (When m is even.)
(11, 10)	$X11Z \xrightarrow{i(\text{even}) \text{ steps}} X'11Z \rightarrow X'10Z \xrightarrow{m-i \text{ steps}} Y10Z$ (With $i \geq 0$.)
(11, 11)	$X11Z \xrightarrow{m \text{ steps}} Y01Z \rightarrow Y11Z$ (When m is odd.) $X11Z \xrightarrow{m \text{ steps}} Y11Z$ (When m is even.)

Lemma 4 Given two distinct vertices $u = X\alpha_2\alpha_1Z$ and $v = Y\beta_2\beta_1Z$ where $(\alpha_2\alpha_1, \beta_2\beta_1)$ is the right most double-bits pair belongs to NEP and $|Z| = 2k \geq 2$ of CQ_n . If there exists a path $P = \langle u, P_1, w, (w)^i, P_2, v \rangle$ for some $0 \leq i \leq 2k-1$, then $\text{len}(P) \geq d_{CQ_n}(u, v) + 2$.

Proof: Let $d_{CQ_n}(u, v) = m$. Suppose $\text{len}(P) \leq d_{CQ_n}(u, v) + 1 = m + 1$. Then $\text{len}(P_1) + \text{len}(P_2) \leq m$. Hence, $2k \leq j \leq n-1$ for each edge $(y, (y)^j)$ of the paths P_1 and P_2 . (If not, $\text{len}(P_1) + \text{len}(P_2) > m$). Therefore, $(w)^i_i = v_i \neq u_i$. We get a contradiction. Thus, $\text{len}(P) \geq d_{CQ_n}(u, v) + 2$. This complete the proof. $\diamond$

Theorem 1 Given two distinct vertices $u = X\alpha_2\alpha_1Z$ and $v = Y\beta_2\beta_1Z$ of CQ_n such that $(\alpha_2\alpha_1, \beta_2\beta_1)$ is the right most double-bits pair belong to NEP, $|Z| = 2k$, and $|X| = |Y| = l = n - 2(k+1)$. There exists the path of length $d_{CQ_n}(u, v) + 1$ if and only if one of the following condition exists.

- (1) $d_{CQ_l}(X, Y) \geq 1$ is odd and $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 01), (00, 10), (00, 11), (01, 00), (01, 01), (01, 10), (10, 00), (10, 01), (10, 11), (11, 00), (11, 10), (11, 11)\}$,
- (2) $d_{CQ_l}(X, Y) \geq 2$ is even and $(\alpha_2\alpha_1, \beta_2\beta_1) \in$

$\{(00, 01), (00, 10), (00, 11), (01, 00), (01, 10), (01, 11), (10, 00), (10, 01), (10, 11), (11, 00), (11, 01), (11, 10)\}$,
 (3) $d_{CQ_l}(X, Y) = 0$, $|X| = |Y| \neq 0$, and $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 11), (01, 10), (10, 01), (11, 00)\}$.

Proof: To prove this theorem, we give an construction scheme for each condition. For the cases not listed in the conditions, we prove there is no such path of length $d_{CQ_n}(u, v) + 1$.

Case 1: $d_{CQ_l}(X, Y) \geq 1$ is odd. Let $m = d_{CQ_n}(X, Y)$.

Subcase 1.1: $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 01), (00, 10), (00, 11), (01, 00), (01, 01), (01, 10), (10, 00), (10, 01), (10, 11), (11, 00), (11, 10), (11, 11)\}$.

Without loss of generality, we can assume that $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 01), (00, 10), (00, 11), (01, 01), (01, 10), (10, 11), (11, 11)\}$. We list the construct scheme for construction the path of length $d_{CQ_n}(u, v) + 1$ in the Table 2.

Based on the K. Efe's shortest path routing algorithm [6], the length of shortest path in this case is $m + 1$. On our scheme, we can create a path of length $m + 2$. Hence, there exists the path of length $d_{CQ_n}(u, v) + 1$ in this case.

Table 2: The construct scheme for the path of length $d(u, v)_{CQ_n} + 1$ in Subcase 1.1.

$(\alpha_2, \alpha_1, \beta_2, \beta_1)$	The path of length $d_{CQ_n}(u, v) + 1$.
(00, 01)	$X00Z \xrightarrow{m-1 \text{ steps}} X'00Z \rightarrow X'01Z \rightarrow Y11Z \rightarrow Y01Z$
(00, 10)	$X00Z \xrightarrow{m-1 \text{ steps}} X'00Z \rightarrow X'01Z \rightarrow Y11Z \rightarrow Y10Z$
(00, 11)	$X00Z \xrightarrow{m \text{ steps}} Y00Z \rightarrow Y10Z \rightarrow Y11Z$
(01, 01)	$X01Z \rightarrow X00Z \xrightarrow{m \text{ steps}} Y00Z \rightarrow Y01Z$
(01, 10)	$X01Z \rightarrow X00Z \rightarrow X10Z \xrightarrow{m \text{ steps}} Y10Z$
(10, 11)	$X10Z \rightarrow X11Z \xrightarrow{m \text{ steps}} Y01Z \rightarrow Y11Z$
(11, 11)	$X11Z \rightarrow X10Z \xrightarrow{m \text{ steps}} Y10Z \rightarrow Y11Z$

Subcase 1.2: $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(01, 11), (11, 01)\}$. Without loss of generality, we can assume that $(\alpha_2\alpha_1, \beta_2\beta_1) = (01, 11)$. Clearly, $d_{CQ_n}(u, v) = d_{CQ_i}(X, Y) = m$ is odd in this case, and $2k+2 \leq j \leq n-1$ for each edge $(x, (x)^j)$ of the shortest path joining u and v . Suppose there exists a path P joining u and v with the length equal to $d_{CQ_n}(u, v) + 1 = m + 1$. Hence, there exists at least one edge, say $(w, (w)^i)$, of P not in a shortest path joining u and v . Therefore, we can write P as $\langle u, P_1, w, (w)^i, P_2, v \rangle$. Clearly, $\text{len}(P_1) + \text{len}(P_2) = m$, and $2k+2 \leq j \leq n-1$ for each edge $(y, (y)^j)$ of the paths P_1 and P_2 . Then we divide the subcases by i .

Subcase 1.2.1: $0 \leq i \leq 2k-1$. By Lemma 4, $\text{len}(P) \geq d_{CQ_n}(u, v) + 2$. We get a contradiction.

Subcase 1.2.2: $i = 2k$ or $2k+1$. Suppose that $\text{len}(P_1)$ is even. Hence, $w_{2k+1}w_{2k} = 01$. Suppose that $i = 2k$. Then $(w)_{2k+1}^i(w)_{2k}^i = 00$. Clearly $(w)_{2k+1}^i(w)_{2k}^i = v_{2k+1}v_{2k} = 00$. We get a contradiction. Suppose that $i = 2k+1$. Then $(w)_{2k+1}^i(w)_{2k}^i = 11$. Since $\text{len}(P_1)$ is even, $\text{len}(P_2)$ is odd. Therefore, $v_{2k+1}v_{2k} = 01$. We get a contradiction.

Suppose that $\text{len}(P_1)$ is odd. Hence, $w_{2k+1}w_{2k} = 11$. Suppose that $i = 2k$. Then $(w)_{2k+1}^i(w)_{2k}^i = 10$. Thus, $v_{2k+1}v_{2k} = 10$. We get a contradiction. Suppose that $i = 2k+1$. Then $(w)_{2k+1}^i(w)_{2k}^i = 01$. Since $\text{len}(P_1)$ is odd, $\text{len}(P_2)$ is even. Therefore, $v_{2k+1}v_{2k} = 01$. We get a contradiction.

Subcase 1.2.3: $2k+2 \leq i \leq n$. Clearly, $\text{len}(P) = m+1$ is even, and all the edges, say $(y, (y)^i)$, in P are with $2k+2 \leq i \leq n-1$. Hence, $v_{2k+1}v_{2k} = 01$. We get a contradiction.

With these three subcases, it is easily to know there is no such path P with $\text{len}(P) = d_{CQ_n}(u, v) + 1$ in this case.

Case 2: $d_{CQ_i}(X, Y) \geq 2$ is even. Let $m = d_{CQ_i}(X, Y)$.

Subcase 2.1: $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 01), (00, 10), (00, 11), (01, 00), (01, 10), (01, 11), (10, 00), (10, 01), (10, 11), (11, 00), (11, 01), (11, 10)\}$. Without loss of generality, we can assume that $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 01), (00, 10), (00, 11), (01, 10), (01, 11), (10, 11)\}$. In this case, we try to add one more step in the routing from X to Y . If it works, then we can construct the path of length $d_{CQ_n}(u, v) + 1$ in Table 3.

Clearly, these paths are all of length $d_{CQ_n}(u, v) + 1$.

Subcase 2.2: $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(01, 01), (11, 11)\}$. Without loss of generality, we can assume that $(\alpha_2\alpha_1, \beta_2\beta_1) = (01, 01)$. Clearly, $d_{CQ_n}(u, v) = d_{CQ_i}(X, Y) = m$ is even in this case, and all the edges, say $(x, (x)^j)$, of the shortest path are with $n \geq j \geq 2k+2$. Suppose there exists a path P joining u and v with length equal to $d_{CQ_n}(u, v) + 1 = m + 1$. Hence, there exists at least one edge, say $(w, (w)^i)$, of P not in a shortest path joining u and v . Without loss of generality, we can write P as $\langle u, P_1, w, (w)^i, P_2, v \rangle$. Clearly, $\text{len}(P_1) + \text{len}(P_2) = m$, and each edge, say $(y, (y)^j)$, of P_1 or P_2 is with $2k \leq j \leq n-1$. Then we divide the subcases by i .

Subcase 2.2.1: $0 \leq i \leq 2k-1$. By Lemma 4, $\text{len}(P) \geq d_{CQ_n}(u, v) + 2$. We get a contradiction.

Subcase 2.2.2: $i = 2k$ or $2k+1$. Suppose that $\text{len}(P_1)$ is even. Hence, $w_{2k+1}w_{2k} = 01$. Suppose that $i = 2k$. Then $(w)_{2k+1}^i(w)_{2k}^i = 00$. Thus, $v_{2k+1}v_{2k} = 00$. We get a contradiction. Suppose that $i = 2k+1$. Then $(w)_{2k+1}^i(w)_{2k}^i = 11$. Since $\text{len}(P_1)$ is even, $\text{len}(P_2)$ is even. Therefore, $v_{2k+1}v_{2k} = 11$. We get a contradiction.

Suppose that $\text{len}(P_1)$ is odd. Hence, $w_{2k+1}w_{2k} = 11$. Suppose that $i = 2k$. Then

Table 3: The construct scheme for the path of length $d(u, v)_{CQ_n} + 1$ in Subcase 2.1.

$(\alpha_2\alpha_1, \beta_2\beta_1)$	The path of length $d_{CQ_n}(u, v) + 1$.
(00, 01)	$X00Z \rightarrow X10Z \xrightarrow{m-1 \text{ steps}} Y'10Z \rightarrow Y'11Z \rightarrow Y01Z$
(00, 10)	$X00Z \rightarrow X01Z \xrightarrow{m-1 \text{ steps}} Y'11Z \rightarrow Y'10Z \rightarrow Y10Z$
(00, 11)	$X00Z \xrightarrow{m \text{ steps}} Y00Z \rightarrow Y10Z \rightarrow Y11Z$
(01, 10)	$X01Z \rightarrow X00Z \rightarrow X10Z \xrightarrow{m \text{ steps}} Y10Z$
(01, 11)	$X01Z \rightarrow X'11Z \rightarrow X'10Z \xrightarrow{m-1 \text{ steps}} Y10Z \rightarrow Y11Z$
(10, 11)	$X10Z \rightarrow X00Z \xrightarrow{m-1 \text{ steps}} Y'00Z \rightarrow Y'01Z \rightarrow Y11Z$

$(w)_{2k+1}^i(w)_{2k}^i = 10$. Thus, $v_{2k+1}v_{2k} = 10$. We get a contradiction. Suppose that $i = 2k + 1$. Then $(w)_{2k+1}^i(w)_{2k}^i = 01$. Since $\text{len}(P_1)$ is odd, $\text{len}(P_2)$ is odd. Therefore, $v_{2k+1}v_{2k} = 11$. We get a contradiction.

Subcase 2.2.3: $2k+2 \leq i \leq n$. Clearly, $\text{len}(P) = m+1$ is even, and all the edges, say $(y, (y)^i)$, in P are with $2k+2 \leq i \leq n-1$. Hence, $v_{2k+1}v_{2k} = 01$. We get a contradiction.

With these three cases, it is easily to know there is no such path P with $\text{len}(P) = d_{CQ_n}(u, v) + 1$ in this case.

Case 3: $d_{CQ_l}(X, Y) = 0$.

Subcase 3.1: $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 11), (01, 10), (10, 01), (11, 00)\}$. Hence, $d_{CQ_n}(u, v) = 2$. Without loss of generality, we may assume that $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 11), (01, 10)\}$. Suppose that $|X| = |Y| \geq 1$. We list the routing scheme in Table 4.

Suppose that $|X| = |Y| = 0$. Then the shortest paths are $\langle u = 00Z, 01Z, 11Z = v \rangle$, $\langle 00Z, 10Z, 11Z \rangle$, $\langle 01Z, 00Z, 10Z \rangle$, or $\langle 01Z, 11Z, 10Z \rangle$. Suppose there exists such path P joining u and v with $\text{len}(P) = 3$ in this case. Then at least one edge, say $(w, (w)^i)$, in P not in a shortest path. Clearly, $0 \leq i \leq 2k-1$. If not, $(w, (w)^i)$ must in a shortest path. By Lemma 4, $\text{len}(P) \geq 4$. We get a contradiction. Thus the path P doesn't exist in this case.

Subcase 3.2: $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(00, 01), (00, 10), (01, 00), (01, 11), (10, 00), (10, 11), (11, 01), (11, 10)\}$. Obviously, $d_{CQ_n}(u, v) = 1$. Since CQ_n has no triangle, there is no such path of length 2 joining u and v in this case.

Subcase 3.3: $(\alpha_2\alpha_1, \beta_2\beta_1) \in \{(01, 01), (11, 11)\}$. Hence, $u = v$. we don't need to consider this case.

This complete the proof. $\diamond$

4 Conclusion

In this paper, we give a scheme to construct the path of length $d_{CQ_n}(u, v) + 1$ joining any two distinct vertices u and v of CQ_n if the path exists or else we prove that the path doesn't exist. The Crossed Cube is a important variation of hypercube Q_n and preserves many of its desirable properties. Thought this paper, the Crossed Cube can be embedded almost any size of path between any two distinct vertices.

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Table 4: The construct scheme for the path of length $d(u, v)_{CQ_n} + 1$ in Subcase 3.1.

$(\alpha_2\alpha_1, \beta_2\beta_1)$	The scheme to create the path joining u and v of length 3
(00, 11)	$X00Z \rightarrow X'00Z \rightarrow X'01Z \rightarrow X11Z$
(01, 10)	$X01Z \rightarrow X'11Z \rightarrow X'10Z \rightarrow X10Z$

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